

Stefano Baldan

Staff Software Engineer · Accessibility, Frontend & Digital Signal Processing

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<https://stefanobaldan.com> · <https://github.com/singintime>

PROFILE

Staff-level software engineer with more than ten years of professional experience in frontend development and Digital Signal Processing. Over four years of leading technically complex accessibility engineering for a large-scale HTML Canvas-based product, from architecture and design through implementation, including the use of AI and LLMs to help hundreds of developers write accessible-by-design code. Individual contributor with a focus on architecture, technical strategy, informal leadership, and multiplying the impact of other engineers. Background in music, sound synthesis and multimedia computing.

WORK EXPERIENCE

Staff Software Engineer · Miro Apr 2024 – present

- Led accessibility engineering for Miro's HTML Canvas-based product, driven by WCAG requirements, EU accessibility regulations, and the goal of reducing manually identified accessibility issues.
- Used AI and LLM tooling to help hundreds of developers produce accessible-by-design code and to scale accessibility checks across the codebase.
- Documented the accessibility architecture through ADRs, RFCs and technical documentation, enabling wider implementation across teams.

Frontend Software Engineer · Miro Nov 2021 – Mar 2024

- Research and development of accessibility solutions for custom web content rendered in the HTML Canvas, personally driving the work from design through implementation.
- Designed and built keyboard navigation across board objects, including navigation logic for free-form, non-grid Canvas layouts (nearest-object selection in four directions) and semantic object tab order.
- Developed a DOM overlay that lets voice-control users interact with Canvas content without significantly affecting browser performance.
- Built alternative views for text and images, with semantic positioning of board content.
- Improved rendering performance in the Miro client using the Canvas and WebGL APIs.

Engineering Team Lead · byrd Jan 2021 – Oct 2021

Logistics company — systems for shipping parcels from origin to destination

- Lead and management of a full-stack development team of 4 people.
- Frontend development in Angular and TypeScript.
- Backend microservices built with Python, Flask, and SQLAlchemy.

Senior Frontend Engineer · byrd Nov 2019 – Dec 2020

- Developed a single-page application for the company's Warehouse Management System.
- Built reactive pages and UI components in Angular and TypeScript.

Frontend Developer · Contralto Audio Sep 2017 – Dec 2019

- Developed a 3D editor for the acoustic simulation of loudspeaker positioning in live venues.
- Created custom 3D graphics components and GPU shaders on top of Three.js and WebGL, shipped as a desktop application with Angular and Electron.

Research Assistant · Iuav University of Venice Jan 2014 – Dec 2017

- Research for the EU project SkAT-VG (Sketching Audio Technologies using Vocalizations and Gestures): <https://skatvg.unipa.it/>
- Leader of the Work Package on imitation-driven sound synthesis.
- Developed physically-informed sound synthesis and analysis algorithms in Python and C/C++.

Freelance IT Consultant · Self-employed Oct 2011 – Dec 2013

- Web-based edutainment games for Pearson Italia; interactive media and sound design for digital art installations.

TECHNICAL SKILLS

Accessibility: WCAG, keyboard and spatial navigation, voice-control support, DOM overlays for Canvas content, accessible-by-design practices

Frontend: TypeScript, JavaScript, Angular, HTML Canvas, WebGL, Three.js, Node.js, Electron

AI: LLM-based tooling for code generation and automated accessibility checks

Backend & other: Python, Flask, SQLAlchemy, C/C++

DSP & audio: Digital signal processing, physically-informed sound synthesis and analysis, audio and multimedia computing

Engineering practice: Architecture and technical strategy, ADRs, RFCs, technical documentation

EDUCATION

MSc in Economics · University Niccolò Cusano

2017 – 2020

Main subjects: Environmental Economics, Statistics, Accounting. Final grade: 110/110.

BSc & MSc in Computer Science for New Media Communications · University of Milan

2005 – 2011

Main subjects: Software Engineering, Digital Signal Processing, Distributed Systems, Human-Computer Interaction, Music Information Retrieval, Semiotics. Final grade: 110/110 cum laude.

RELEVANT PROJECTS

- **Audio plugins** — independent line of audio plugins built on physically and statistically grounded DSP (parametric and analytical methods), distributed via Polar.sh. <https://stefanobaldan.com/plugins/>
- **vodit** — minimalist dependency injector. No frills, no dependencies, no decorators. <https://singintime.github.io/vodit>
- **SDT (Sound Design Toolkit)** — open-source C library for physically-based sound synthesis; developer and maintainer, 2014–2017. <https://github.com/SkATVG/SDT>
- **Mals** — performance presented in Milan in 2013. Co-author. <https://vimeo.com/86330894>
- **Sonic Tennis** — audio-only game for iOS devices. <https://youtu.be/0uxqhCS0GPg>
- **GlobeMX** — interactive HTML5 media player. Master's thesis project. <http://emipiu.di.unimi.it>
- **IPCapture** — Processing library for acquiring MJPEG live streams. <https://github.com/singintime/ipcapture>
- **Musica Sull'Acqua** — sonification of the civic aquarium of Milan. <https://ludovico.lim.di.unimi.it/download/papers/SMC2012.pdf>
- **Moonlanding** — interactive installation, Festival della Creatività, Florence, 2009. Sound designer. <https://youtu.be/kEERdTvCbYI>

LANGUAGES

Italian (native) · English (C2) · French (C1) · German (B1)

OTHER

People person, passionate student, and lifelong learner. Linux user and open-source enthusiast. Digital artist and maker. Pianist, mandolinist and singer with good reading and playing skills and a solid base in musical theory. Tracking, mixing and mastering engineer for several home-recording projects. Amateur traditional folk dancer, juggler, and fencer.